

# TorqBall — Boost & Attachments

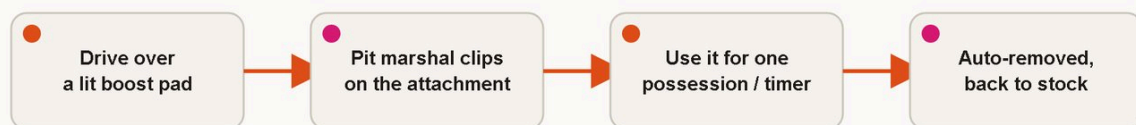
2026-09-06. Full build spec for TorqBall's own boost system — the answer to "Rocket League has boost, our cars can't fly or safely run raw video-game speed." Locked in [[TorqBall — Design Study v1]]: a boost changes the **car** (a quick-swap strap-on attachment), the **ball/goal** (a field effect), or uses a **safe speed route** (a fleet-wide turbo toggle only — never a per-car speed hack, and never a "lane," see §2 Turbo). Every car-attachment straps direct over the stock bumper (v1 mount, §1) — velcro/zip-tie only, **nothing bolts, screws or glues to a rented car**. Boost pads reuse the tag-per-car + zone-reader tech being built for [[TorqCapture — Zone-Capture Tech]]. Pairs with [[TorqBall — Rulebook]] (§6, where this plays in-match) and [[TorqBall — Stretch Modes]] (Rumble = this system turned to 11).

## The rule, stated once

### TorqBall — boost system

Earn on a pad → claim at the pit → use it → lose it • never a per-car speed hack

TORQWARS



#### The set

|         |                                                  |
|---------|--------------------------------------------------|
| Plow    | wide bumper — move ball + shove                  |
| Scoop   | curved lip — lift/carry the ball                 |
| Grabber | velcro face — the ball sticks, you dribble       |
| Turbo   | speed burst via a boost lane / fleet toggle only |
| Magnet  | the goal you attack widens ~20 s                 |

**Nothing bolts to a rented car. No per-car speed hack, ever.** Every boost below is either a strap-on attachment (car), a change to the ball/goal (field), or the fleet-wide turbo toggle (speed) — never a modification that makes one specific car go faster than the fleet's safety cap.

## 1. The universal mount (reused, not re-invented)

**Mount system — v1 is strap-direct**, same call as [[TorqSiege — What The Event Needs]] §2: every attachment velcro-cinches straight over the stock bumper with 2 straps + foam padding underneath (the

straps ARE the mount), straps off clean in seconds at handback. On rented-fleet cars this is the only justified v1 mount — TorqSiege's own needs doc explicitly downgrades the printed **mount plate + dovetail rail** to "an owned-fleet upgrade, not an event-#1/#2 cost," and TorqBall rents from the same fleet, so the same logic applies here: no v1 event pays for the rail.

- **v1 (rented fleet, every event):** Plow/Scoop/Grabber heads strap direct over the stock bumper/padded plate — no rail, no per-car printed part beyond the head itself.
- **Owned-fleet upgrade (future, not costed into any event #1/#2):** a printed mount plate carrying a dovetail rail, so an earned boost swaps the head in seconds without re-strapping. Worth building once TorqWars owns cars outright; until then it stays a convenience on the roadmap, not a line item.

## 2. The five boosts

| Boost          | Effect                                                                     | How it's built                                                                                               | Type | Duration/earn                                                                                                                                 |
|----------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Plow</b>    | Wide bumper — moves the ball more decisively + bulldozes rival cars aside  | Strap-on paddle, plywood or PLA print, straps direct over the stock bumper (v1 — no rail; §1)                | Car  | One possession or until bumped off (marshal call)                                                                                             |
| <b>Scoop</b>   | Curved lip — lifts/carries the ball a little, feeds the Hoops stretch mode | Strap-on curved front piece, PLA or bent plywood                                                             | Car  | One possession or until bumped off                                                                                                            |
| <b>Grabber</b> | Velcro face — the ball sticks so you can dribble it until bumped off       | Velcro-faced plow attachment + a matching velcro patch stitched/glued onto the practice/event ball's surface | Car  | Until bumped off by a rival, or a hard timer (~15–20 s) if it proves too dominant in playtesting — <b>gated by the bench test below (§2a)</b> |

| Boost         | Effect                                                                                                                                                                                                                                                   | How it's built                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Type                                                 | Duration/earn                                     |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|
| <b>Turbo</b>  | A real speed burst                                                                                                                                                                                                                                       | <b>Sole mechanism: a fleet-wide "turbo" toggle</b> — a TX/app throttle-limit raise applied to every car on the fleet simultaneously for a short window, then reverted. <b>Feasibility gate, unconfirmed:</b> ask the rental club at the car-deal stage whether their fleet's TX or app supports a global throttle-limit change. If yes → this is the whole build (₹0 hardware — a settings routine the tech runs). If no → <b>Turbo is SHELVED for v1</b> (dropped from the earn set; Rumble and the other four boosts run without it) — there is no fallback lane, because a low-friction or sloped strip does not deliver speed (friction is what lets a tyre grip and accelerate; a slicker surface makes a capped car slide/spin, not go faster) and the venue is a flat indoor hall with no real downhill. A low-friction patch remains a legitimate <i>chaos</i> feature on its own terms — see [[TorqBall — Stretch Modes]] §4 Hazard/Arena Variant — but it is never sold here as a Turbo speed mechanism | Speed (safe, fleet-wide only — never a per-car hack) | ~5–8 s toggle window, called/reverted by the tech |
| <b>Magnet</b> | <b>The goal mouth you're attacking widens ~20 s</b> — not your own goal; rewards pushing forward, never a defensive relief. (Reconciled 2026-09-06 — [[TorqBall — Design Study v1]] originally stated the opposite direction; see that doc's changelog.) | <b>Marshal-actuated field effect, not a mechanism cut into the boards.</b> A lightweight foam marker/panel that a marshal manually slides across the attacking team's goal mouth on a call, visibly widening the target for ~20 s — the same category of thing as a dead-ball call, not an automated track/rail cut into a rigid goal opening.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Field (goal)                                         | ~20 s, then the marshal slides it back            |

## 2a. Grabber bench test (added 2026-09-06 — gates the build)

Grabber is the one boost that is a genuinely untested physical unknown, on the same footing as the ball and the boards: whether velcro reliably grabs on a glancing hit, releases cleanly on a rival's bump, and whether a capped ~1 kg car can controllably drag a stuck 40–60 cm ball (especially a light inflatable, which may deform or peel the patch) is not knowable from a spec sheet. **Full protocol lives in [[TorqBall — Ball & Boards Test]] §Grabber Bench Test** (added alongside this fix) — summary here: with the winning ball + a capped car + a velcro-patched shell, measure (1) grab-on-contact rate across ~10 approach angles, (2) controllable dribble distance/turn radius, (3) clean-release rate on a rival's bump, (4) patch durability over ~10 hits. **PASS** = grabs on a clear majority of contacts, dribbles a useful distance/turn, releases cleanly most of the time, patch survives 10 hits → build the Grabber head as speced above. **CONDITIONAL** = works but needs a constraint (e.g. the hard ~15–20 s release timer, a reinforced patch) → buildable with that constraint. **FAIL** = doesn't grab reliably, or drags uncontrollably, or the patch fails within 10 hits → drop

Grabber from the v1 earn set until the mechanism is reworked (different velcro grade, smaller patch, a lighter head).

### Notes per boost

- **Plow / Scoop** are the safe defaults — pure geometry changes, nothing electronic, cheapest to build and hardest to break.
- **Grabber is the standout mechanic** — it's the one that makes dribbling (not just shoving) possible, which is a genuinely new skill layer over plain push-football. It needs a **velcro patch on the ball itself**, so the event ball (and its spares, per [[TorqBall — Ball & Boards Test]]) must be prepped with a loop-side velcro patch stitched or glued to its shell before the event — do this to every spare ball, not just the match ball, so a mid-match ball swap doesn't kill the mechanic. Build only after §2a's bench test clears.
- **Turbo is the one boost that touches speed**, and the only build that is speed-safe: a fleet-wide toggle that raises every car's throttle limit for a short window, called and reverted by the tech — never a lane, never a per-car change. If the club's fleet can't support the toggle, Turbo simply isn't built for that event; there is no fallback mechanism, by design, because every fallback considered (a slick lane, a sloped ramp) either doesn't work physically or reintroduces an unsafe per-car hack.
- **Magnet** is the only boost that changes the *attacking* team's own opportunity rather than a rival's car — it rewards good field position (driving the ball toward the widened goal you're shooting at) rather than raw contact. **It's marshal-actuated, plainly, like a dead-ball call** — a marshal slides the marker/panel across the goal mouth by hand and slides it back after ~20 s. There is no track, rail or mechanism cut into the board goal, so no test/rig gates it the way Grabber's velcro does (§2a).

## 3. Boost pads — the earn mechanism

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Lit floor pads placed inside the boarded pitch. Driving over one earns a boost, which is then claimed and clipped on at the pit lane (not instantaneously mid-play — this keeps the attach/detach step safe and off the moving pitch).

- **Tech reuse:** the same **tag-per-car + zone-reader pipeline** being built for [[TorqCapture — Zone-Capture Tech]] is the pad detector — a boost pad is just a tiny zone (the same overhead-cam or UHF-tag pipeline, sized down from a ~1.5–2 m capture ring to a ~30–50 cm pad). Car-over-pad auto-lights, logs the earn to the controller, and the MC calls it live off the same scoreboard feed TorqCapture uses.
- **v1 fallback (no electronics needed to launch):** painted pads (coloured gaffer squares), called by a marshal or read off the overhead content cam — the same fallback logic as TorqCapture's flat-taped rings. Nothing stops TorqBall running boost pads on day one with zero electronics spend.
- **Which mode runs pads:** per [[TorqBall — Rulebook]] §6, Standard and Duel run boost-free by default (keeps the base game legible for a first bracket); Big Party and the Rumble stretch mode turn pads on.

## 4. Small BOM

| Item                                                          | Qty — why                                                             | Est Rs                                                      | Notes                                                                                                                                                                               |
|---------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Padded strap-mount plate (per fitted car, v1 — no rail; §1)   | 6–8 — sized to a Big Party/Rumble session, not the whole fleet        | 0 (plywood offcut/3D-printed, filament cost only)           | Straps direct over the stock bumper; the printed dovetail rail is an owned-fleet upgrade, not costed here (§1)                                                                      |
| Velcro straps + foam padding (per mount)                      | 6–8 sets                                                              | 600–1,200 (lot)                                             | Same strap stock priced in the shared fleet-protection doc                                                                                                                          |
| Plow head (PLA/plywood)                                       | 2–3                                                                   | 300–800 (lot)                                               | Cheapest attachment, print or cut in-house                                                                                                                                          |
| Scoop head (PLA/bent plywood)                                 | 2–3                                                                   | 300–800 (lot)                                               |                                                                                                                                                                                     |
| Grabber head (velcro-faced plow)                              | 2–3                                                                   | 400–1,000 (lot)                                             | Includes the hook-side velcro on the head; build only after the §2a bench test PASSes                                                                                               |
| Velcro loop patches for the ball(s)                           | 4–6 (every match ball + spare, per [[TorqBall — Ball & Boards Test]]) | 100–300 (lot)                                               | Stitched or glued to the ball shell                                                                                                                                                 |
| Grabber bench-test kit (pre-event, one-off)                   | 1 practice ball + patch set                                           | 300–600                                                     | Consumed running the §2a test before any Grabber head is built for an event                                                                                                         |
| Magnet goal-widener marker/panel (marshal-actuated, no track) | 1 per goal, 2 total                                                   | 800–2,000                                                   | A lightweight foam panel/marker board the marshal slides by hand across the goal mouth on a call — no rail, drawer-slide or automated mechanism, so there's no rig to build or test |
| Boost pads — painted fallback (v1)                            | 4–6                                                                   | 200–500 (lot)                                               | Coloured gaffer tape squares                                                                                                                                                        |
| Boost pads — electronic (reuses TorqCapture rig, if built)    | shares the Zone-Capture camera/controller                             | 0 (incremental — the rig is already bought for TorqCapture) | Only a marginal software/marker cost once the TorqCapture rig exists; see that doc's BOM for the base spend                                                                         |
| Turbo (fleet-toggle mechanism)                                | —                                                                     | 0                                                           | No hardware — a TX/app setting, if the club's fleet supports it (§2 feasibility gate); if not, shelved for v1 at zero cost either way                                               |

| Item                                                       | Qty — why | Est Rs        | Notes                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------|-----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One-time boost-kit subtotal (car + field, v1 painted pads) |           | ≈ 3,000–7,200 | Rows: 0+600+300+300+400+100+300+800+200+0+0 = 3,000 low; 0+1,200+800+800+1,000+300+600+2,000+500+0+0 = 7,200 high. Excludes the shared TorqCapture electronics (a separate cross-game investment, not re-bought here). No "boost lane" line — Turbo dropped the lane mechanism entirely (§2); the Hazard/Arena Variant's own low-friction patch is costed in [[TorqBall — Stretch Modes]], not here |

## 5. Safety statement

- **Nothing bolts, screws or glues to a rented car** — every attachment is velcro/zip-tie strap-on, comes off clean at handback, same rule as the foam bump-rings and the TorqSiege weapon mounts.
- **No unsafe per-car speed hack** — Turbo is fleet-wide only (a toggle applied and reverted by the tech to every car equally, and only if the club's fleet supports it — otherwise shelved), never a single car's throttle unlocked beyond the safety cap, and never delivered via a low-friction "lane" (which reduces grip, not increases speed).
- **Attach/detach happens at the pit lane, not mid-play** — a boost is claimed off the pad, then clipped on between possessions, keeping the fiddly strap-on step away from a moving scrum.
- **Magnet and boost pads are field/software effects** — no moving hazard is introduced into the pitch itself; the goal-widener panel slides on a fixed track, not into the play area.
- **Grabber ships only after its bench test (§2a) clears** — an untested velcro-dribble mechanic doesn't go into a live scrum on say-so, same standard as the ball and the boards.

**Verified + changelog (2026-09-06):** first build from the locked boost system in [[TorqBall — Design Study v1]]. Specced all five boosts (Plow/Scoop/Grabber/Turbo/Magnet) with effect, build method, type (car/field/speed) and earn/duration rules; confirmed every car-attachment reuses the TorqSiege universal strap-on mount + dovetail rail, and that Turbo is strictly environmental/fleet-wide, never a per-car speed hack. Specced the Grabber ball-velcro-patch requirement across every match ball and spare. Wrote the boost-pad mechanism (tag+zone-reader reuse from TorqCapture, CV/UHF pipeline sized down to a small pad, with a zero-electronics painted-pad+marshal/cam fallback for v1). Costed a small BOM (~Rs 4,400–11,600 one-time for the car+field kit at v1 painted-pad scope, explicitly excluding the shared TorqCapture electronics spend). Stated the safety rule (nothing bolted, no unsafe speed hack, attach/detach at the pit lane only) plainly.

**Second pass (2026-09-06, examiner fixes):** (1) **Magnet direction reconciled** — restated as "the goal you're attacking widens," matching the corrected [[TorqBall — Design Study v1]] (previously said the opposite — "your" goal widens — a direct contradiction this doc's original wording did NOT have, but the Design Study and Rulebook needed to catch up to it; see both docs' changelogs). (2) **Turbo rebuilt around a single verified-feasible mechanism** — dropped the "boost lane = speed" claim entirely (a low-friction/sloped strip reduces traction, it does not increase speed, and there's no real downhill on a flat indoor floor); Turbo is now solely the fleet-wide TX/app throttle-toggle, gated on club confirmation, with an explicit SHELVE-if-unconfirmed fallback instead of an invented lane mechanism. (3) **Added §2a — a Grabber bench test** (grab-on-contact rate, dribble control, clean-release, patch durability over 10 hits) with a

PASS/CONDITIONAL/FAIL bar, gating the Grabber build the same way the ball and boards are gated; full protocol cross-referenced in [[TorqBall — Ball & Boards Test]]. (4) **Mount system reframed to strap-direct v1** — matching [[TorqSiege — What The Event Needs]] §2's logic (rented fleet  $\Rightarrow$  no printed rail as an event cost; it's an owned-fleet upgrade only). (5) **BOM recomputed**: removed the boost-lane row (₹1,000–3,000, no longer a Turbo cost), added a Grabber bench-test consumables row (₹300–600), and reconciled the subtotal arithmetic — old figure (₹4,400–11,600) didn't sum to its own rows (high end overstated by ₹500 against a ₹0 mount-plate line); new subtotal  $\approx$  **₹3,700–8,700**, shown with the row-by-row sum inline so it reproduces.

**Third pass (2026-09-06, examiner fix): Magnet downgraded from an ungated novel mechanism to a marshal-actuated field effect.** §2's Magnet row previously described "a marker slides out from the goal frame on a track/rail" — an automated mechanism cut into the rigid board goal that, unlike Grabber, was never bench-tested (§2a exists for Grabber only). Reworded Magnet to a **marshal-actuated** panel/marker a marshal manually slides across the attacking team's goal mouth for ~20 s, explicitly the same category as a dead-ball call — stated plainly so it's clear no test/rig is needed (matching the "Notes per boost" paragraph, now updated to say the same). §4's BOM line renamed "Magnet goal-widener marker/panel (marshal-actuated, no track)" and repriced **₹1,500–3,500  $\rightarrow$  ₹800–2,000** (a simple hand-slid foam panel needs no rail/drawer-slide hardware); one-time boost-kit subtotal recomputed **₹3,700–8,700  $\rightarrow$  ₹3,000–7,200** (row-by-row sum shown inline). This lower figure feeds [[TorqBall — What The Event Needs]] §3/§5, which is corrected in that doc's own changelog to match.